MIL-C-48080 (MU)
31 July 1972
SUPERSEDING
PA-PD-3130
4 December 1970

MILITARY SPECIFICATION

CARTRIDGE, 152MM, TP-T, XM411E6
LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This Military Specification covers the loading, assembling and packing for one type of target practice cartridge designated as Cartridge, 152MM, TP-T, XM411E6 and applies only to engineering test quantities.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-223 - Powder, Black
MIL-A-2550 - Ammunition and Special Weapons, General Specification for
MIL-I-45607 - Inspection Equipment, Supply and Maintenance of
MIL-I-50867 - Ignition Element, Electric; Loading, Assembling and Packaging

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-414 - Sampling Procedures and Tables for Inspection by Variables for Percent Defective
MIL-STD-286 - Propellants, Solid: Sampling, Examination and Testing
MIL-STD-1168- Lot Numbering of Ammunition
MIL-STD-1235- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

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DRAWINGS

ARMY

9212118 - Box, Packing, Ammunition, for Cartridge, 152MM, HEAT-T-MP, M409

9212119 - Container, Ammunition, Fiber, M556A1, for Cartridge, 152MM, HEAT-T-MP, M409

9212218 - Igniter Assembly

9260121 - Cartridge Assembly with Bag, Barrier, Elastomeric

9249028 - Ignition Element, Electric

9254948 - Cartridge, 152MM, TP-T, XM411E6

PUBLICATIONS

ARMY

ET-9254948- Index of Inspection Equipment Lists, for Cartridge, 152MM, TP-T, XM411E6

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 General.-Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Cartridge,-The cartridge shall comply with all requirements specified on Drawing (dwg.) 9254948 and with all requirements specified in applicable specifications.

3.3 Assembly Torque.-Thread joints shall withstand subsequent reapplication of the minimum (min.) assembly torque specified on the applicable drawing, without moving, while thread sealing compounds, (if applicable), are wet. (See 6.5).

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3.4 Moisture Content of Black Powder in Igniter Assemblies.-The moisture content of the black powder, loaded into igniter bags at time of loading, shall not exceed 0.70 percent (%) when determined in accordance with MIL-P-223.

3.5 Workmanship

3.5.1 All parts shall be free of chips, dirt, grease, rust, burrs, cracks and other foreign material. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents.

3.5.2 The painting and marking shall be restored by the assembling contractor on any components furnished to him painted and marked, when such paintings have been marred or obliterated. (Bare spots in painting, other than marking, to total not more than 1/4 square inch in aggregate, are allowable.)

3.6 Ignition Element Circuit Continuity.-The resistance of the ignition element after insertion into the cartridge base shall not be greater than 1.0 ohm when tested in accordance with 4.3.3.

3.7 Drying Time.-After assembly of cartridge case base to cartridge case body, the cartridge shall be cured in the conditioning bay in accordance with the drying time specified on the applicable drawing.

3.8 Moisture Content of the Cartridge Case.-At the time of assembly of elastomeric barrier bag to the cartridge, the average moisture content of the case body and case base shall not exceed 1.5%, and no individual reading shall exceed 2.0% when tested in accordance with 4.2.3.7.

3.9 Pre-Use Inspection of Cartridge Case Components.-The gap between the cartridge case body outside diameter and the cartridge case base inside diameter shall not exceed .010 inch for a depth of .0625 inch (max.) when inspected in accordance with 4.3.5.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of Product.-At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with an identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

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4.2 Quality Conformance Inspection

4.2.1 Lot Formation.-The term "inspection lot", as used in this specification, is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions, and complies with the provisions for submission of product as specified in Standard MIL-STD-105. Changes to the process, specifications or drawings, not affecting safety, performance, interchangeability or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each Inspection lot of loaded complete round cartridges shall contain:

a. Ignition element assemblies of one lot interfix number from one manufacturer.

b. Cartridge case components (bodies and bases) of one lot interfix number from one manufacturer.

c. Projectile metal parts of one lot interfix number from one manufacturer.

d. Propellant charges of one lot from one manufacturer.

e. Black powder of one lot from one manufacturer.

f. Tracers of one lot interfix number from one manufacturer.

g. Cartridge case locking rings produced to the same drawing, drawing revisions, specification and specification revision by one manufacturer.

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h. Plug and disc assemblies produced to the same drawing, drawing revisions, specification and specification revision by one manufacturer.

i. Cartridge case mounting rings produced to the same drawings, drawing revision, specification and specification revision by one manufacturer.

4.2.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (See 4.2.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for Critical Defects (and Major Defects, when so specified) shall be 100 percent. Sampling plans for Major and Minor Defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of Major and Minor Defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each Major Defect and an AQL of 0.65 percent for each Minor Defect. When this method is elected, it shall be applied to all of the Major and Minor Defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 Projectile Loading Assembly (See dwg. 9254948).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 Percent		
101.	Tracer missing or inverted	Visual	01001
102.	Plug and disc assembly fails to meet min. assembly torque (See 3.3 and 4.3.1)	Test	01002
103.	Thread sealant missing from plug and disc assembly threads of base plug	Test	01003
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (See 3.5)	Visual	01004

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4.2.2.2 Assembly of Ignition Element (See dwg. 9254948).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Ignition element inserted inversely..	Test	02001
Major:	AQL 0.65 Percent		
101.	Ignition element loose.....	Manual	02002
102.	Ignition element below flush tolerance.....	Gage	02003
103.	Ignition element above flush.....	Straight-Edge	02004
104.	Adhesive missing from outside diameter of ignition element or inside diameter of hole in base (prior to assembly).....	Visual	02005
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship.....	Visual	02006

4.2.2.3 Igniter Assembly, Prior to Loading of Black Powder (See dwg. 9212218).

Categories	Defects	Method of Inspection	Code No
Critical: None defined.			
Major:	AQL 0.40 Percent		
101.	Incorrect weight of igniter charge...	Gage	03001
102.	Igniter bag torn, ripped or cut.....	Visual	03002
Minor: None defined.			

4.2.2.4 Igniter Assembly (See dwg. 9254948).

Categories	Defects	Method of Inspection	Code No
Critical: None defined.			

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Major:	AQL 0.65 Percent		
101.	Tube cracked or broken	Visual	04001
102.	Igniter bag charge missing or inverted	Visual	04002
103.	Closing plug missing	Visual	04003
Minor:	AQL 1.0 Percent		
201.	Adhesive applied incorrectly to tube.	Visual	04004
202.	Evidence of poor workmanship	Visual	04005

4.2.2.5 Assembly, After Assembling Cartridge Case (See dwg. 9254948).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	100 Percent Inspection		
101.	Thread sealant missing from case locking ring threads	Visual	05001
102.	Case mounting ring not firmly seated	Manual	05002
103.	Incorrect number of case mounting rings assembled	Visual-Manual	05003
Minor:	None defined.		

4.2.2.6 Assembly, Prior to Assembly of Cartridge Case Base and Prop Charge (See dwg. 9254948).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	100 Percent Inspection		
101.	Case locking ring fails torquing requirements (See 6.5)	Visual	06001
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship	Visual	06002

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4.2.2.7 Assembly (See dwg. 9254948).

Categories	Defects	Method of Inspection	Code No
Critical:	None defined.		
Major:	100 Percent Inspection		
101.	Incomplete, insufficient or incorrect application of adhesive to base (prior to assembly)	Visual	07001
102.	Propelling charge missing (prior to assembly of base)	Visual	07002
103.	Excess adhesive not removed after assembly of base	Visual	07003
104.	Improper or incomplete coating of lubrication on rotating band or lubricant not dry to touch	Visual-Manual	07004
105.	Closing torque on clamp (See 6.6)	Visual	07005
106.	Rotating band damaged	Visual	07006
Major:	AQL 0.40 Percent		
107.	Cartridge case sidewall-base gap incorrect	Gage	07007
108.	Assembly time (See dwg. 9254948)	Test	07008
Minor:	AQL 1.50 Percent		
201.	Marking incorrect or unidentifiable including color coding	Visual	07009
202.	Evidence of poor workmanship	Visual	07010
203.	Cartridge case-projectile gap incorrect	Visual	07011

4.2.2.8 Unsealed Fiber Container, Prior to Loading of Cartridge (See dwg. 9212119).

Categories	Defects	Method of Inspection	Code No
Critical:	None defined.		

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Categories	Defects	Method of Inspection	Code No
Major:	100 Percent Inspection		
101.	Barrier bag (elastomeric) torn, ripped or missing from cartridge.....	Visual	08001
102.	Neck tube or ring missing.....	Visual	08002
103.	Desiccant bag broken, missing or not stitched into ring.....	Visual	08003
104.	Sleeve (two halves) missing or improperly assembled.....	Visual	08004
Major:	AQL 0.65 Percent		
105.	Round cannot be removed by hand (after inserting).....	Manual	08005
106.	Cushion missing.....	Visual	08006
107.	Discs between cushion and cartridge missing.....	Visual	08007
108.	Nose support missing or inverted.....	Visual	08008
Minor:	None defined.		

4.2.2.9 Sealed Fiber Container (See dwg. 9212119).

Categories	Defects	Method of Inspection	Code No
Critical:	None defined.		
Major:	AQL 0.65 Percent		
101.	Container damaged or cut through all impregnated layers.....	Visual	09001
102.	Sealing strip applied improperly.....	Visual	09002
103.	Metal end loose or distorted.....	Visual-Manual	09003
Minor:	AQL 2.5 Percent		
201.	Contents loose.....	Manual	09004
202.	Tear tab length inadequate.....	Visual	09005
203.	Gap between body and cover in excess of 1/8 inch.....	Visual	09006
204.	Marking misleading or unidentifiable.	Visual	09007
205.	Evidence of poor workmanship.....	Visual	09008

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4.2.2.10 Sealed Wooden Packing Box (See dwg. 9212118).

Categories	Defects	Method of Inspection	Code No
Critical:	None defined.		
Major:	100 Percent Inspection		
101.	Barrier bag improperly sealed, torn, or missing (prior to sealing box)....	Visual	10001
102.	Contents loose (prior to sealing)....	Manual	10002
Major:	AQL 0.65 Percent		
102.	Hardware or strapping missing, broken, loose or improperly located.....	Visual-Manual	10003
103.	Box damaged, exposing contents.....	Visual	10004
104.	DOD code marking missing, incorrect or unidentifiable.....	Visual	10005
Minor:	AQL 0.65 Percent		
201.	Car seal missing, unsealed or improperly positioned.....	Visual	10006

4.2.3 Testing

4.2.3.1 Check Test for Deterioration of Ignition Elements - Major Defect - Code No. 05009.-If the total elapsed time between original acceptance of an assembled ignition element lot and assembly of that lot into the ammunition components with which it is to be issued exceeds two (2) years, or if the ignition elements have been subjected to adverse conditions at any time since previous tests, or if more than six (6) months have elapsed since a previous check test, the ignition element lots shall be subjected to, and must satisfactorily pass, the check test for deterioration specified in MIL-I-50867, prior to use. (See 6.4).

4.2.3.2 Moisture Content of Black Powder (See 3.4) - Major Defect - Code No. 05010.-The contractor shall provide adequate controls to assure the compliance of each lot with the applicable requirements. For verification, and just prior to the start of each eight (8) hours loading production, the contractor shall test one sample of black powder selected from that quantity, or sub-lot of powder which is be used in the eight-hour loading production of ignition element base sub-assemblies. Composite samples shall not be used. Testing shall be in accordance with

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MIL-P-223, and if the moisture content of the sample exceeds the requirement and loading has not begun, the sub-lot of powder represented by the sample shall be rejected. If powder from a sub-lot with excessive moisture has been used in loading, the remaining unloaded powder and all assemblies containing the non-conforming powder, shall be rejected.

4.2.3.3 Profile and Alignment (See dwg. 9254948) - Major Defect - Code No. 10009.-The cartridges shall be gaged 100% for profile and alignment and the maximum allowable force for extracting the round from the chamber shall be applied. Any cartridge which fails to comply with the applicable requirement shall be classed defective and removed from the lot. The test shall be performed using equipment in accordance with 4.2.4.

4.2.3.4 Pull Test of Cartridge Case Base (See dwg. 9254948). - Major Defect - Code No. 10010.-This test shall be conducted on a 100% basis. Any cartridge which fails to comply with the applicable requirement shall be removed from the lot. Testing shall be as specified in 4.3.2.

4.2.3.5 Ignition Element Circuit Continuity (See 3.6) - Major Defect - Code No. 15001.-A sample shall be randomly selected from each shift's production of the ignition element-base subassemblies which are scheduled for use in a complete round lot. The sample size shall be determined from MIL-STD-105, General Inspection Level II. The samples shall be tested in accordance with 4.3.3, using equipment in accordance with 4.2.4. Failure of any sample to comply with the applicable requirement shall require 100% testing of the ignition element-base subassemblies represented by the failing sample. Any subassembly failing to comply with the requirements shall be rejected and removed from the shift's production.

4.2.3.6 Drying Time - Major Defect - Code No. 05006 (100% Inspection).-Each skid of assembled cartridges shall be visually inspected in accordance with 4.3.4 for drying time in the conditioning bay. If any skid document indicates time in the conditioning bay lower than the minimum total requirement, the skid shall be placed back in the conditioning bay until compliance with the applicable requirement is indicated (See 6.7).

4.2.3.7 Moisture Content of Cartridge Case (See 3.8) - Major Defect - Code No. 20001.-Just prior to removing each skid of cartridges from the conditioning bay after final drying, the relative humidity (RH) and the temperature in the conditioning bay shall be checked and recorded on the skid document (See 6.7). If the RH is above the maximum allowable level, or if the temperature does not comply wit

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the applicable requirement, no cartridges will be removed from the conditioning bay. Instead, the RH and/or the temperature in the conditioning bay must be corrected to reach an acceptable level, and the moisture test specified in MIL-STD-286B, Method 101.4, must be performed on one cartridge selected at random from the skid which was scheduled for removal. This cartridge will represent all cartridges in the conditioning bay at the time the discrepancy was observed. The samples required for the test shall be taken without removing the selected cartridge from the conditioning bay and shall be prepared as follows:

Two (2) .025 gm (approximate) samples shall be sectioned from the case body at random locations, and two (2) .025 gm (approximate) samples shall be sectioned from the case base. The temperature control shall be set at 100°C with a timer setting of 15 minutes. Cartridges represented by the sample shall remain identified until successful performance during the moisture content test is indicated. Acceptable performance during the moisture test will qualify the skid for removal from the conditioning bay and shall reinstate normal inspection for RH and Temperature. Under no circumstances will any cartridges in the conditioning bay be removed until acceptable performance during the moisture test is indicated. If the samples fail to comply with the applicable requirement, all cartridges in the conditioning bay at the time the discrepant RH and/or Temperature readings were observed will remain in the conditioning bay an additional two (2) hours + 1/2 hour (in addition to the final curing time specified on the drawing). One (1) hour prior to completion of the additional curing time, samples shall again be taken from the cartridge originally selected and subjected to testing as specified above. Failure of these samples to comply with the applicable requirement shall reject the skid originally scheduled for removal and will require moisture testing as above on one cartridge selected at random from each subsequent skid of cartridges until acceptable performance is demonstrated. Skids indicating non-conformance shall also be rejected. Normal inspection may be reinstated coincident with determination of acceptable moisture content.

4.2.3.8 Pre-Use Inspection of Cartridge Case Components - Major Defect - Code No. 20003.-Just prior to the start of each production lot, fifteen (15) cartridge case bodies and bases shall be randomly selected from the cases and bases scheduled for use in the complete round lot. The samples shall be inspected in accordance with 4.3.5. Failure of any sample shall require flaring of the failing sample(s) and reinspection. If reinspection indicates compliance with the applicable requirement (See 3.9), the remainder of the cases and bases represented by the samples shall be flared prior to use. If reinspection still indicates rejection, the cases and bases represented by the sample shall not be used in loading.

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4.2.4 Inspection Equipment.-Equipment Tabulation Number ET-9254948 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3.

4.2.4.1 Government Rights to Documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.2.4.2 Supply and Maintenance.-Supply and maintenance of the equipment listed in the ET shall be in accordance with Specification MIL-I-45607.

4.2.4.3 Government Use of Contractor's Inspection Equipment.-The contractor shall make available to the Government all inspection equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment, and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.3 Test Methods and Procedures

4.3.1 Torque Testing.-Torque shall be applied in a tightening direction while holding the parts immediately adjacent to the joints being tested so as to avoid possible loosening of any other joints. The joint under test shall be marked to provide a means for detection of any movement between the adjacent parts. Any discernible movement between adjacent parts shall be regarded as a torque test failure.

4.3.2 Pull Test of Cartridge Case Base.-This test shall be performed after the bond at the cartridge case body-base joint has had sufficient time to set. The cartridge shall be held in a suitable test fixture clamping on the projectile body. The test apparatus shall apply a longitudinal load of 150 pounds distributed over 360° around the lip of the cartridge case base. Observation shall be made for separation of the cartridge case base from the cartridge. The method of test should not damage the ammunition components, and paint or markings damaged during the test shall be restored. Cartridges which meet the requirement may be returned to the lot provided they have not been damaged by the method of test.

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4.3.3 Ignition Element Circuit Continuity.-The ignition element assembly, after insertion into the cartridge case base, shall be tested for resistance by the following method: A probe shall be inserted through the test holes of the base and ignition element to the inner sleeve of the ignition element and the circuit completed by connecting a lead from the probe to the outer cup of the ignition element. An ohmmeter shall be attached and the resistance read to the nearest 0.1 ohm. No more than 10 milli-amperes current shall be induced in the circuit during this test.

4.3.4 Drying Time.-For each skid of cartridges, the document representing the skid shall be annotated with the time and date tha the skid of cartridges entered the conditioning bay and the time and date that the skid of cartridges was removed from the condi-tioning bay.

4.3.5 Pre-Use Inspection of Cartridge Case Components.-The cartridge case base shall be assembled to the cartridge case body (without glue) and the base clamp assembled and torqued to the required value. A .010 inch thick by .375 inch wide feeler gage shall be used to check the gap between the outside diameter of the cartridge case body and the inside diameter of the cartridge case base. The gap shall be checked at four locations, 90° apart. The gage shall not enter to a depth exceeding .0625 inch at any point.

5. PREPARATION FOR DELIVERY

5.1 Level A.-The cartridge shall be preserved and packaged in accordance with dwg. 9212119. In addition, the container and desiccant bag (cover removed from body with one each desiccant bag placed inside the body) shall be dried in a circulating oven for a min. of 8 hours at a temperature of 100°F + 10°F, 10% RH max., prior to packing. If the container is not used within thirty minutes after removal from the oven, the container shall be closed tightly, with desiccant bag remaining inside. Final packing of the cartridge in the container shall be accomplished within eight hours of the time the container was removed from the oven.

5.2 Packing

5.2.1 Level A.-The containers shall be packed in accordance with dwg. 9212118.

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6. NOTES

6.1 Ordering Data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data Cards - Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

6.2 Inspection Code Numbers.-The five-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. These code numbers are used to correlate the characteristics cited on the Equipment Lists with the inspections listed in this specification. They should also be cited as references on drawings of equipment designs submitted by the contractor (See 6.3.5).

6.3 Inspection Equipment.-The contractor shall design inspection equipment as required by the inspection Equipment Lists (EL) referenced on the applicable ET in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Inspection Equipment Lists (EL).-Inspection Equipment Lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipme. (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes, as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Inspection Equipment List Codes.-The inspection equipment, as defined in 6.3.3 and 6.3.4, will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.

MU - Army design, Mandatory for Use

6.3.3 Army Designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed

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for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor-prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor Designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which completely depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category A, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit one copy of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of Contractor Designs.-All submitted designs shall contain a reference to the applicable five-digit code number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as Critical and Major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-QA-A-A. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged; however, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

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6.4 Cost of Check Test.-The contracting officer will arrange for the contractor to be reimbursed for the expense incurred in the performance of check tests. The test shall be conducted at Government expense without cost to the contractor who loaded the primers or to the contractor assembling the cartridge, and shall not constitute a basis for rejection against either contractor, except where deterioration has occurred as a direct result of carelessness in handling, storage, etc., permitted while the primer lot was under the jurisdiction of either contractor.

6.5 Torquing of Case Locking Ring.-Due to the compressibility of the cartridge case material during the assembly of the case body to the projectile, the case locking ring will lose its torque almost immediately. Inspection for compliance with the applicable torquing requirement is therefore limited to witnessing the torquing operation.

6.6 Torquing of Base Clamp.-Due to the requirement prohibiting reapplication of assembly of torque, inspection for compliance with the applicable torquing requirement is limited to witnessing the torquing operation.

6.7 Skid Documents.-Documentation on relative humidity and temperature in the conditioning bay, as well as drying time, shall be retained for each skid until the lot which the documents represent has been finally accepted.

6.8 Gluer Qualification.-Since the gluing operation associated with the assembly of the cartridge case base to the cartridge case body is extremely important to overall cartridge case strength, it is recommended that personnel associated with this operation be carefully trained. A gluer qualification test is suggested. Properly assembled cartridge case body to base joints should be capable of withstanding 1000 pounds of force during pull testing.

Custodian:
Army-MU

Preparing Activity:
Army-MU

Project Number: 1320-A-140

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